

BREADTH-FIRST SEARCH (BFS) ROUTE ANALYSIS

Home at Street 03, PAF Falcon Complex → Federal Urdu University, Gulshan Campus

Origin	Home - Street 03, PAF Falcon Complex (near The City School PAF Chapter)
Destination	Federal Urdu University of Arts, Sciences & Technology - Gulshan Campus, Block 9, University Road

1. Problem Statement

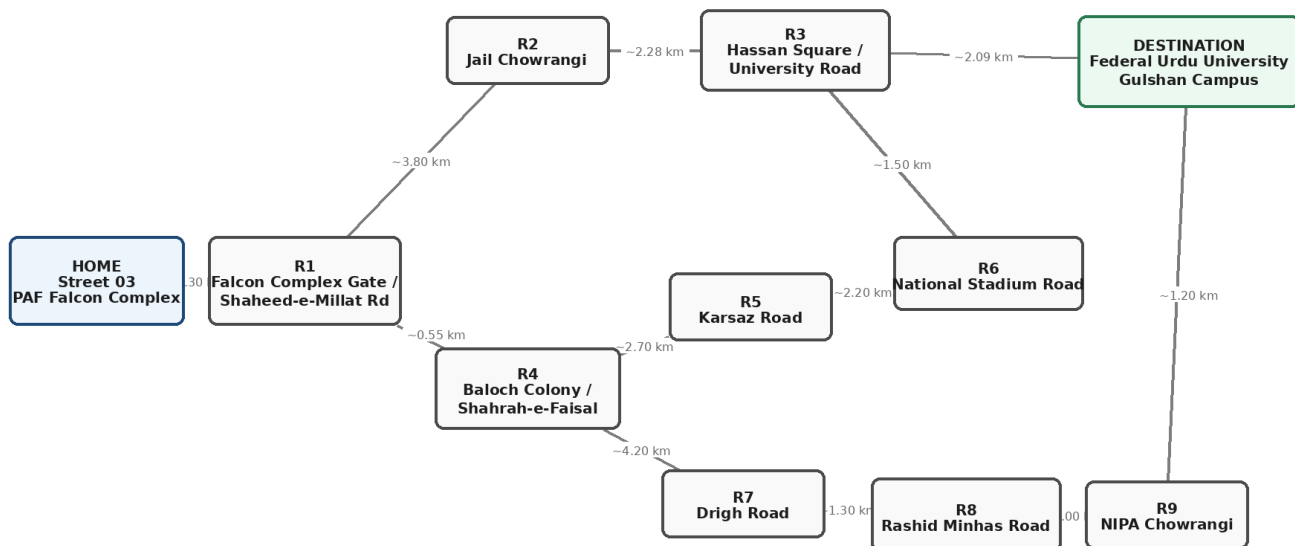
The task is to model realistic road connections from the home location inside PAF Falcon Complex to Federal Urdu University (Gulshan Campus) and apply Breadth-First Search (BFS). BFS identifies the path with the fewest graph edges, while the weighted comparison shows how route distance can differ from hop count.

The corrected graph begins from Falcon Complex itself. It uses Shaheed-e-Millat Road as the main direct approach toward Jail Chowrangi and University Road, plus two genuine alternative corridors through Karsaz / National Stadium Road and Drigh Road / Rashid Minhas Road / NIPA.

2. Corrected Route Graph

Corrected Route Graph: Home to Federal Urdu University

Starting point: Street 03, PAF Falcon Complex (near The City School PAF Chapter)



Distances marked ~ are approximate academic edge weights, not live-navigation measurements.

Figure 1. Simple box-and-line graph using roads connected to the actual home area

3. Adjacency List

Node	Connects to
Home	R1
R1 - Falcon Complex Gate / Shaheed-e-Millat Rd	Home, R2, R4
R2 - Jail Chowrangi	R1, R3
R3 - Hassan Square / University Road	R2, R6, Federal Urdu University
R4 - Baloch Colony / Shahrah-e-Faisal	R1, R5, R7
R5 - Karsaz Road	R4, R6
R6 - National Stadium Road	R5, R3
R7 - Drigh Road	R4, R8
R8 - Rashid Minhas Road	R7, R9
R9 - NIPA Chowrangi	R8, Federal Urdu University
Federal Urdu University	R3, R9 - destination

4. BFS Trace

Step	Action	Queue after step
Init	Q = [Home]	V = [Home]
S1	Dequeue Home → discover R1	Q = [R1]
S2	Dequeue R1 → discover R2, R4	Q = [R2, R4]
S3	Dequeue R2 → discover R3	Q = [R4, R3]
S4	Dequeue R4 → discover R5, R7	Q = [R3, R5, R7]
S5	Dequeue R3 → R2 visited; discover R6 and Federal Urdu University	Q = [R5, R7, R6, FUUAST]
S6	Dequeue R5 → R4 visited; R6 already discovered	Q = [R7, R6, FUUAST]
S7	Dequeue R7 → discover R8	Q = [R6, FUUAST, R8]
S8	Dequeue R6 → R5 and R3 already visited	Q = [FUUAST, R8]
S9	Dequeue FUUAST → destination reached; stop	Done

BFS destination discovery: The destination is first discovered when R3 (Hassan Square / University Road) is expanded. The parent chain gives Home → R1 → R2 → R3 → Federal Urdu University.

5. Route Comparison

Route	Path	Edges	Calculation (approx.)	Distance
Route A - BFS route	Home → R1 → R2 → R3 → FUUAST	4	$0.30 + 3.80 + 2.28 + 2.09$	8.47 km
Route B - Karsaz / Stadium	Home → R1 → R4 → R5 → R6 → R3 → FUUAST	6	$0.30 + 0.55 + 2.70 + 2.20 + 1.50 + 2.09$	9.34 km
Route C - Drigh / NIPA	Home → R1 → R4 → R7 → R8 → R9 → FUUAST	6	$0.30 + 0.55 + 4.20 + 1.30 + 3.00 + 1.20$	10.55 km

6. Results and Interpretation

- BFS shortest path (fewest edges): Home → Falcon Complex Gate / Shaheed-e-Millat Road → Jail Chowrangi → Hassan Square / University Road → Federal Urdu University.
- This corrected BFS path is based on roads that actually connect to the Falcon Complex / Shaheed-e-Millat area; Johar Chowrangi and Balochistan Sajji are no longer used as starting-area nodes.
- With the approximate academic weights used in this graph, Route A is also the shortest of the three compared routes by distance.
- Route B gives a realistic alternative through Shahrah-e-Faisal, Karsaz and National Stadium Road before joining University Road.
- Route C gives another realistic alternative through Drigh Road, Rashid Minhas Road and NIPA before approaching the university from University Road.

7. Route Accuracy Note

Public location references place PAF Falcon Complex / The City School PAF Chapter on Shaheed-e-Millat Road and Federal Urdu University's Gulshan Campus on Main University Road, Block 9. Public road guides also show that Jail Chowrangi connects Shaheed-e-Millat Road to University Road, and that Federal Urdu University lies on the University Road corridor beyond Hassan Square.

The edge distances marked with “~” are approximate academic weights for the graph, not live Google Maps or GPS measurements. Traffic restrictions, U-turns, entry gates, and the exact house position within Street 03 can change the real driving distance. The graph structure and road names are the important part for the BFS assignment.

8. Final Answer

Correct BFS path: Home (Street 03, PAF Falcon Complex) → Falcon Complex Gate / Shaheed-e-Millat Road → Jail Chowrangi → Hassan Square / University Road → Federal Urdu University, Gulshan Campus.